

3. We return to the assumptions on σ and F with more details in the next subsection. See Remarks & Examples 3.1 in [MMM11] for a discussion of assumptions about the cross section, and Lemma 6.1 in the same reference for the proof of statements in some cases.

Formally, passing to the limit when $\varepsilon \rightarrow 0$ in equation (1.1), we obtain that the limit f^0 is in the kernel of Q which is spanned by the equilibrium F , which means that $f^0 = \rho(t, x)F(v)$. Thus, it amounts to find the equation satisfied by the density ρ . This question of approximation of kinetic equations by macroscopic equations has a long history, dating back to the pioneering works of E. Wigner [WB61], A. Bensoussan et al. [BLP79], as well as E.W. Larsen and J.B. Keller [LK74]. Since then, numerous papers have addressed this topic (for further references, see the papers by C. Bardos et al. [BSS84], and P. Degond et al. [DGP00]). The resulting equations fall into two categories, depending on the rate of decrease in velocity of the equilibrium F . It is well-known (see [DGP00] for instance) that when F decreases *rapidly* for large velocities (such as when F follows a Maxwellian distribution function), the distribution function f^ε converges to $\rho(t, x)F(v)$ as ε goes to 0, for the classical scaling $\theta(\varepsilon) = \varepsilon^2$, with ρ being the solution of the diffusion equation

$$\partial_t \rho - \nabla_x \cdot (D \nabla_x \rho) = 0, \quad (1.5)$$

where

$$D = \int_{\mathbb{R}^d} (v \otimes v) \frac{F(v)}{\nu(v)} dv.$$

When F decreases *slowly* and it is a heavy tail distribution function, satisfying

$$F(v) \sim \frac{\kappa}{|v|^{d+\alpha}} \quad \text{as } |v| \rightarrow \infty \quad (1.6)$$

for some $\alpha > 0$, the previous diffusion matrix D , given by (1.5) might be infinite. In that case, the diffusion limit leading to (1.5) breaks down, which means that the choice of time scale $\theta(\varepsilon) = \varepsilon^2$ was inappropriate. It has been shown in [MMM11] and [Mel10], using different methods and under various assumptions, that in such cases, the appropriate time scale involves the parameter α , which appears in the equilibrium. More specifically, for $\theta(\varepsilon) = \varepsilon^\gamma$ with γ depends on α , the following fractional diffusion equation is obtained

$$\partial_t \rho + \kappa (-\Delta_x)^{\frac{\gamma}{2}} \rho = 0, \quad (1.7)$$

where the fractional Laplacian appearing in the previous equation can be defined by the following singular integral

$$(-\Delta_x)^{\frac{\gamma}{2}} u(x) := c_{\alpha, d} \text{PV} \int_{\mathbb{R}^d} \frac{u(x) - u(y)}{|x - y|^{d+\gamma}} dy.$$

Let us now discuss in more detail the contexts of the last two references cited. In [MMM11], the authors addressed the problem in the space homogeneous case (that is with σ independent of x). They proved that when F satisfies (1.6) and the collision frequency ν satisfies

$$\nu(v) \sim \nu_0 |v|^\beta \quad \text{as } |v| \rightarrow \infty,$$

then for $\alpha > 0$ and $\beta < \min(\alpha; 2 - \alpha)$, and by taking $\gamma := \frac{\alpha - \beta}{1 - \beta}$, the scaling $\theta(\varepsilon) = \varepsilon^\gamma$ leads to the previous fractional diffusion equation. While in [Mel10], the same problem has been addressed but for a cross section which depends on the position variable x but assuming that the collision frequency is bounded, more precisely

$$0 < \nu_1 F(v) \leq \sigma(x, v, v') \leq \nu_2 F(v), \quad (1.8)$$

i.e. for $\beta = 0$ compared to the case of [MMM11], which gives in particular,

$$0 < \nu_1 \leq \nu(x, v) \leq \nu_2. \quad (1.9)$$